

Plotting w/ Python

Motivation

Annoying to switch tools

Imagine:

- Download data
- Convert from Excel to csv
- Import in Python & analyze
- Export results to Excel to graph



Repeat every time you screw up, get new data, ...

What if you didn't have to?

Matplotlib to the Rescue!

Open source

Plotting library for visualization in Python

- Bar graphs
- Histograms
- Line graphs
- Scatter Plots
- Pie Charts (for the boss who doesn't know not to use them)

Warning! NumPy Array vs Python List

NumPy array differences:

- Homogeneous
- Fixed data type
- Performance improvements
- Functions for mathematical & other operations

Want details? <https://www.geeksforgeeks.org/python-lists-vs-numpy-arrays/>

```
import numpy as np

a = np.array([1, 2, 3])
print(a)
```

```
[1 2 3]
```

```
import numpy as np

a = np.array([(1, 2, 3), (4, 5, 6)])
print(a)
```

```
[[1 2 3]
 [4 5 6]]
```

But You CAN Use Python Lists For Plotting!

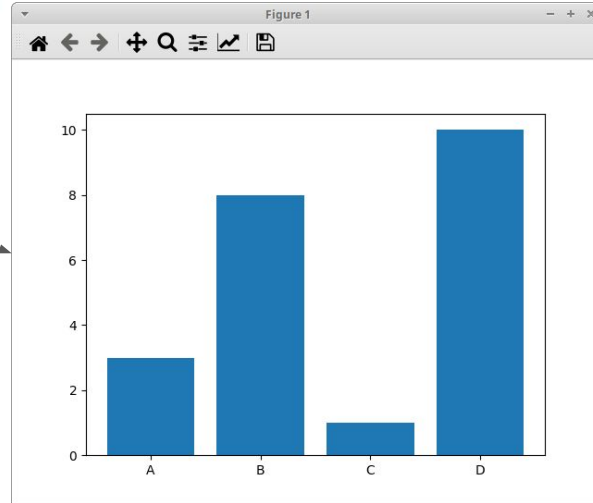
Needs import matplotlib
line

Bar Graph Example

```
testplt.py - /home/toth/
File Edit Format Run Options Window
import numpy as np

x = np.array(["A", "B", "C", "D"])
y = np.array([3, 8, 1, 10])

plt.bar(x,y)
plt.show()
```



```
testplt.py - /home/toth/
File Edit Format Run Options Window
import matplotlib.pyplot as plt

x = ["A", "B", "C", "D"]
y = [3, 8, 1, 10]

plt.bar(x,y)
plt.show()
```

Histogram Example

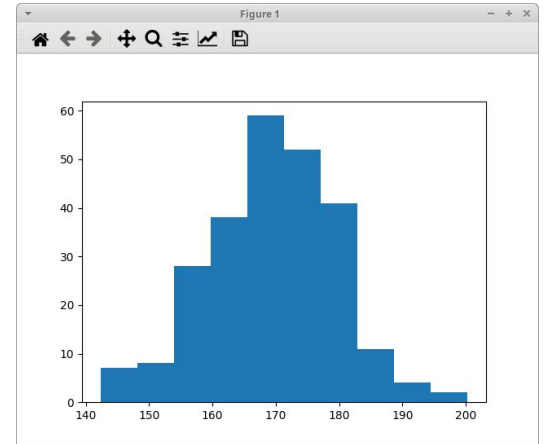
```
testplt.py - /home/toth/Desktop/te
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt
import numpy as np

x = np.random.normal(170, 10, 250)

plt.hist(x)
plt.show()

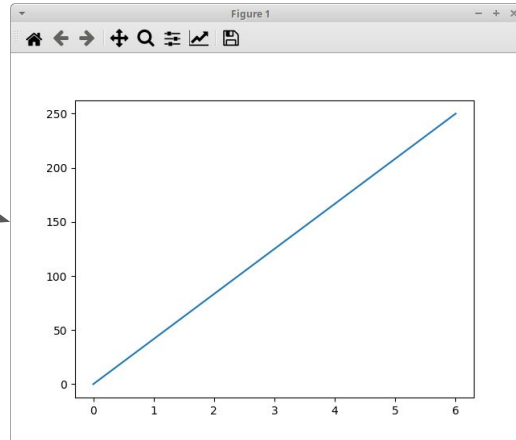
l_data = []
for item in x:
    l_data.append(item)

plt.hist(l_data)
plt.show()
```



Line Graph Example

```
testplt.py - /home/  
File Edit Format Run Options Win  
import matplotlib.pyplot as plt  
import numpy as np  
  
xpoints = np.array([0, 6])  
ypoints = np.array([0, 250])  
  
plt.plot(xpoints, ypoints)  
plt.show()
```



```
testplt.py - /home/to  
File Edit Format Run Options Windo  
import matplotlib.pyplot as plt  
  
xpoints = [0, 6]  
ypoints = [0, 250]  
  
plt.plot(xpoints, ypoints)  
plt.show()
```

Don't need import numpy line

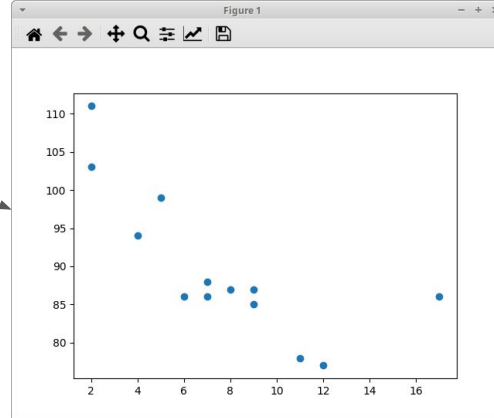
Scatter Plot Example

```
testplt.py - /home/toth/Desktop/testplt.py (3.10.12)  
File Edit Format Run Options Window Help
```

```
import matplotlib.pyplot as plt  
import numpy as np
```

```
x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])  
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
```

```
plt.scatter(x, y)  
plt.show()
```



```
testplt.py - /home/toth/Desktop/testp  
File Edit Format Run Options Window Help
```

```
import matplotlib.pyplot as plt  
import numpy as np
```

```
x = [5,7,8,7,2,17,2,9,4,11,12,9,6]  
y = [99,86,87,88,111,86,103,87,94,78,77,85,86]
```

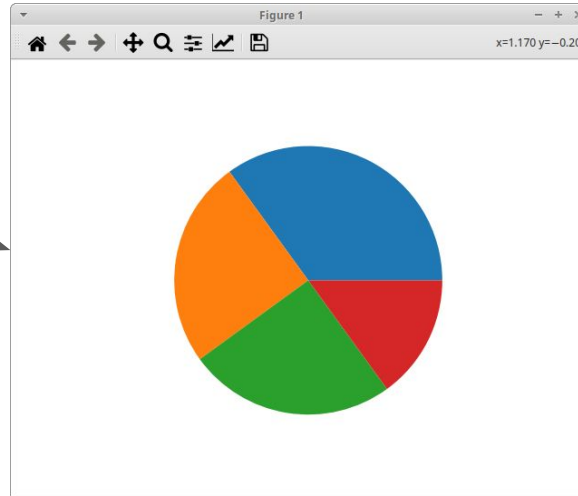
```
plt.scatter(x, y)  
plt.show()
```

Pie Chart Example

```
testplt.py - /home/
File Edit Format Run Options Win
import matplotlib.pyplot as plt
import numpy as np

y = np.array([35, 25, 25, 15])

plt.pie(y)
plt.show()
```



```
testplt.py - /home
File Edit Format Run Options Win
import matplotlib.pyplot as plt

y = [35, 25, 25, 15]

plt.pie(y)
plt.show()
```

Links

Main link for plotting: https://www.w3schools.com/python/matplotlib_intro.asp
(don't worry about this for now)

Basic Plots

Bar Graphs: https://www.w3schools.com/python/matplotlib_bars.asp

Histograms: https://www.w3schools.com/python/matplotlib_histograms.asp

Line Graphs: https://www.w3schools.com/python/matplotlib_pyplot.asp

Scatter Plots: https://www.w3schools.com/python/matplotlib_scatter.asp

Pie Charts: https://www.w3schools.com/python/matplotlib_pie_charts.asp

Making Your Plots Nice

Markers: https://www.w3schools.com/python/matplotlib_markers.asp

Line styles: https://www.w3schools.com/python/matplotlib_line.asp

Axis labels & plot titles: https://www.w3schools.com/python/matplotlib_labels.asp

Grid lines: https://www.w3schools.com/python/matplotlib_grid.asp

Drawing multiple plots in one figure:

https://www.w3schools.com/python/matplotlib_subplot.asp

Lab Time

Take your time with the lab and read the pages linked to slides 12 & 13 to really get a sense of what you can do!

Lab is linked to Moodle